

PROJECT REPORT

EVENTVERSE

An Event and Festival Announcement System in Python

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1. Introduction

Events and festivals such as cultural programs, college fests, fairs, religious celebrations, sports meets and workshops take place all around us. Yet information about them is often scattered across posters, messaging groups and word of mouth. People miss events because they do not know when or where they are happening, and organizers struggle to reach the right audience.

Eventverse is a Python-based project that acts as a single place to announce upcoming events and festival celebrations. It stores and displays the details that people generally need: the name and type of the event, date and timing, venue, organizer name and contact, and the list of participants. The project applies core Python concepts learned in the Python Essential course, namely variables, data types, lists, dictionaries, functions, loops, conditions, file handling and modular programming.

2. Problem Statement

There is no simple, organized system through which a community, college or locality can announce events and let people look up complete information. Announcements are inconsistent, details such as venue, timing or organizer contact are frequently missing, and participant records are kept manually. Eventverse addresses this by providing a structured, searchable and easy-to-use event announcement system.

3. Objectives

- To create a central platform for announcing upcoming events and festivals.
- To store complete organizer details (name and contact) for every event.
- To record venue and timing information clearly.
- To register and view participant details for each event.
- To allow users to search and filter events by name, date, venue or category.
- To practise Python fundamentals through a practical, real-life project.

4. Scope of the Project

Eventverse is designed for small and medium communities such as colleges, societies, housing colonies, clubs and local organizations. It covers the full basic lifecycle of an announcement: adding an event, viewing upcoming events, registering participants, updating or cancelling an event, and searching. The current version is a menu-driven Python application with data saved in a file, and it can be extended into a web or mobile application later.

5. Features and Modules

Module	Description
Event Management	Add, view, update and delete events with title, category (festival, cultural, sports, technical, etc.), description, date and time.
Organizer Details	Stores the organizer's name, phone number and email for each event so that people can contact them.
Venue Details	Stores venue name, address and city of the event or festival.

Module	Description
Timing and Schedule	Stores start date, end date, start time and end time; upcoming events are listed in date order.
Participant Registration	Register participants with name, contact and age group; view participant list and count per event.
Search and Filter	Find events by keyword, date, venue, city or category.
Data Storage	Saves all records in a JSON/CSV file so that data remains available after the program closes.

6. System Requirements

Type	Requirement
Language	Python 3.10 or above
Libraries	Standard library only: json, csv, datetime
Editor / IDE	IDLE, VS Code or PyCharm
Operating System	Windows, Linux or macOS
Hardware	Any computer with 2 GB RAM or more

7. Data Design

Each event is stored as a Python dictionary. All events are kept in a list and saved to a file. The main fields are shown below.

Field	Type	Example
event_id	int	101
title	str	Diwali Cultural Night
category	str	Festival
date	str (YYYY-MM-DD)	2026-11-08
time	str	06:00 PM - 10:00 PM
venue	str	Community Hall, Sector 5
city	str	Bhopal
organizer_name	str	Anita Sharma
organizer_contact	str	98XXXXXX01
participants	list of dict	[[name, contact], ...]

8. System Design and Working

The program follows a simple menu-driven flow. When it starts, it loads saved events from the file. The user then chooses an option from the main menu. Each option is handled by a separate function, which keeps the code organized and easy to maintain. After every change the data is written back to the file.

Main menu options

- Add a new event
- View all upcoming events
- Search events
- Register a participant
- View participants of an event
- Update or cancel an event

- Exit

9. Sample Implementation (Python)

The following code shows the core logic of Eventverse using functions, dictionaries, lists, loops, file handling and formatted output.

```
import json
from datetime import datetime

FILE = "events.json"

def load_events():
    try:
        with open(FILE, "r") as f:
            return json.load(f)
    except FileNotFoundError:
        return []

def save_events(events):
    with open(FILE, "w") as f:
        json.dump(events, f, indent=4)

def add_event(events):
    event = {
        "event_id": len(events) + 101,
        "title": input("Event name: "),
        "category": input("Category (Festival/Cultural/Sports): "),
        "date": input("Date (YYYY-MM-DD): "),
        "time": input("Timing: "),
        "venue": input("Venue: "),
        "city": input("City: "),
        "organizer_name": input("Organizer name: "),
        "organizer_contact": input("Organizer contact: "),
        "participants": [],
    }
    events.append(event)
    save_events(events)
    print("Event added successfully!")

def view_events(events):
    if not events:
        print("No events announced yet.")
        return
    for e in sorted(events, key=lambda x: x["date"]):
        print(f"[{e['event_id']}] {e['title']} ({e['category']})")
        print(f"    Date: {e['date']} Time: {e['time']}")
        print(f"    Venue: {e['venue']}, {e['city']}")
        print(f"    Organizer: {e['organizer_name']} - {e['organizer_contact']}")
        print(f"    Participants: {len(e['participants'])}")

def register_participant(events):
    eid = int(input("Enter event ID: "))
    for e in events:
        if e["event_id"] == eid:
            name = input("Participant name: ")
            contact = input("Contact: ")
            e["participants"].append({"name": name, "contact": contact})
            save_events(events)
            print("Participant registered!")
            return
    print("Event not found.")

def main():
    events = load_events()
```

```
while True:
    print("\n=== EVENTVERSE ===")
    print("1.Add 2.View 3.Register 4.Exit")
    ch = input("Choice: ")
    if ch == "1": add_event(events)
    elif ch == "2": view_events(events)
    elif ch == "3": register_participant(events)
    elif ch == "4": break
    else: print("Invalid choice.")

main()
```

10. Sample Output

```
=== EVENTVERSE ===
1.Add 2.View 3.Register 4.Exit
Choice: 2
[101] Diwali Cultural Night (Festival)
      Date: 2026-11-08 Time: 06:00 PM - 10:00 PM
      Venue: Community Hall, Sector 5, Bhopal
      Organizer: Anita Sharma - 98XXXXXX01
      Participants: 24
[102] Inter-College Football Meet (Sports)
      Date: 2026-11-15 Time: 09:00 AM - 05:00 PM
      Venue: City Sports Ground, Bhopal
      Organizer: Rohit Verma - 97XXXXXX45
      Participants: 40
```

The names and numbers above are dummy data used only for demonstration.

11. Advantages

- Simple, menu-driven and easy for anyone to use.
- All event information is available in one place.
- Data is saved permanently in a file.
- Uses only the Python standard library, so no extra installation is needed.
- Modular functions make the code easy to understand and extend.

12. Limitations

- It is a console application without a graphical interface.
- Data is stored in a local file, so it is not shared online between multiple users.
- No login system, notifications or payment support in the current version.

13. Future Scope

- Convert the project to a web application using Flask or Django, or a GUI using Tkinter.
- Use a database such as SQLite or MySQL for larger data.
- Add user login for organizers and participants.
- Send reminders through email or SMS.
- Add maps for venues, ticketing and online registration.
- Add event photos, feedback and ratings.

14. Conclusion

Eventverse successfully demonstrates how Python can be used to build a practical system for announcing events and festivals with complete details of organizers, venues, timings and participants. The project helped in applying the concepts of Python Essential in a real-world context, including data structures, functions, file handling and program design. With further enhancements, it can grow into a full-fledged event announcement platform for communities and institutions.

15. References

- Python Software Foundation, Python Documentation - <https://docs.python.org/3/>
- Python Essential course study material and class notes.
- W3Schools Python Tutorial - <https://www.w3schools.com/python/>